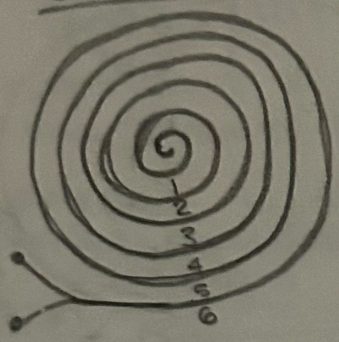
09/23/2026

D_h = 1.8 mm

cellet size (μm)	λ
212	0.118]
300	0.167
350	0.194]
500	0.278

$0.1 \leq \lambda \leq 0.3$ ✓

6 spirals



* don't split the channels into two outlets immediately when spiral ends, give short straight section

experimentally determine how long this section should be.

* start with 10 mm (compare 5mm, 15mm, 20 mm prototypes)

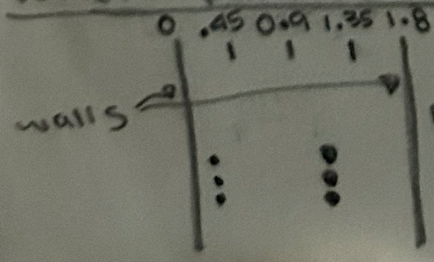
$D_h = \frac{2wh}{w+h} = 1.8$ mm width = 1.8 m, height = 1.8 m

* also provide short straight distance before particles enter spiral to establish more consistent flow (10 mm ?)

* Run 212-300 μm sample alone and see where population focuses, than run 350-500 μm sample. Then determine where to put divider and run mixed sample.

Channels:
- square shape
- can keep same dimensions throughout spirals

Divider: physical wall that separates the two flows into different outlets
experimentally determine where to put this by using microscope to see where particles concentrate



EXAMPLE

around 0.3 mm

around 1.2 mm

put divider between 0.3 - 1.2 mm